T S T R A C K E R

PT3 Song Editor

Compose and edit AY chip-tunes on your own machine

For the Timex Sinclair 2068

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Plays PT2 and PT3 tunes via the PTxPlay (Bulba) driver
Companion to the TS Tracker Player

64K SOFTWARE

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Keep this manual by the keyboard until the key layout is in your fingers.

1. Welcome

TS Tracker turns your Timex Sinclair 2068 into a three-voice music workstation. You can load an existing **PT3** tune from tape and rework it, or start a brand new song from nothing and build it up note by note – no tape required to begin. When you are happy with the result, TS Tracker writes it back out to cassette as a standard song file that the TS Tracker Player (or any PT3 player) can load and play.

The program drives the AY-3-8912 sound chip through the well-known PTxPlay (Bulba) driver, the same engine used by the player, so what you hear while editing is what you will hear on playback.

If you are the sort of person who reads the whole manual before touching the keyboard, good for you – but you needn't. TS Tracker leads you by the hand with on-screen menus and a built-in help page (press K any time you are editing). If you would rather learn by doing, jump to *Walkthrough: your first tune* near the back and follow along; the reference chapters fill in the rest.

2. Loading the program

TS Tracker is supplied as a **.tap** cassette image. Load it the usual way:

```
LOAD ""
```

then start the tape. After a few seconds the **title screen** appears:

```
TS Tracker
PT3 song editor for the TS-2068
a 64K Software production
Press any key to start.
```

Press any key to continue to the first menu.

3. The first screen

When the program starts it has no song in memory, so it shows the **No tape loaded** menu with three choices:

Key	Action
S	**Scan tape** for songs to edit
N	**New song** – start composing with no tape needed
Q	**Quit** back to BASIC

If you just want to make a tune from scratch, press **N** and skip ahead to *The pattern editor* (or to the *Walkthrough*). To work on a song already on cassette, read on.

4. Scanning a tape

Press **S** (or SPACE) and start your song tape playing. TS Tracker reads through the tape and lists every song file it finds, up to nine at a time:

1 (PT3) MY TUNE
2 (PT3) DEMO SONG
3 (PT2) OLD FAVOURITE

Each entry shows its slot number, its format, and its name. When the last song has gone by, press **CAPS SHIFT + SPACE** to stop scanning.

From the directory:

Key	Action
1…9	Choose a song to load and edit
R	Re-scan the tape
Q	Quit

Pick a number and TS Tracker loads that song into memory.

> **Note:** TS Tracker can *edit* PT3 songs. PT2 songs can be loaded and played, but not edited in this version – the editor will tell you so if you try.

5. The song information screen

Whether you loaded a song or started a new one, you arrive at the **Song info** screen. It is the hub you return to from the pattern editor, and it shows the tune's vital statistics:

Field	Meaning
File	The song's name
Type	PT3 (or PT2)
Speed	Playback speed (tempo) value
Positions	Length of the song, and its loop point
Patterns	How many patterns the song uses

From here:

Key	Action
V	Open the pattern editor
CAPS SHIFT + **7** / **6**	Raise / lower the song's **Speed** (tempo)
Q	Go back

About Speed. It is the number of 50ths-of-a-second the chip holds each row, so a *larger* number plays *slower*. Most tunes sit around **3 to 6**. Watch the **Speed** field change as you press CAPS+7 / CAPS+6. The new tempo is stored with the song and takes effect the next time you play.

6. The pattern editor

This is where the work happens. A *pattern* is a short block of up to 64 **rows**, and each row holds one event for each of the three sound channels (**A**, **B** and **C**). A song plays its patterns one after another in the order set by its position list.

The editor shows sixteen rows at a time:

```
Pat 00/00 Row 00 Ch:A Oct:4
RR  ChA s   ChB s   ChC s
00  C-4 .  --- .  --- .
01  --- .  --- .  --- .
02  --- .  --- .  --- .
...
```

- The line at the top tells you the current **pattern number**, the **cursor row**, the active **channel**, and the current **octave**. (The right-hand part of this line doubles as the edit-mode readout – see *Setting volumes, samples and more* below.)
- Each channel cell shows a **note** and a **sample** number. **---** means "nothing here"; **-=-** is a **rest** (stop sounding); **C-4** is the note C in octave 4.
- Every fourth row is tinted **cyan** and every sixteenth **yellow**, so you can find the beat at a glance, like the bar lines on sheet music.
- The cursor highlights one cell. The right-hand **s** column shows that cell's sample.

At the bottom, **Free: NNNNN bytes** tells you how much room is left for the song when it is saved.

6.1 Moving around

Key	Moves
CAPS SHIFT + **7** / **6**	Up / down a row
CAPS SHIFT + **5** / **8**	Left / right between channels A, B, C
R	Jump to row 0 (top of the pattern)

Key	Moves
O / P	Previous / next pattern
F	Jump to a pattern by number (type two hex digits)

A joystick works for up/down/left/right as well.

> **Your edits are never lost.** Unlike many trackers, TS Tracker keeps the whole song decoded in memory while you work. Moving between patterns, playing the song, and saving all happen automatically – there is no separate “commit” step to remember. Just edit and go.

6.2 Entering notes

TS Tracker lays a piano keyboard across two rows of keys. The bottom row gives the natural notes; the row above gives the sharps:

```
sharps:  S  D     G  H  J
notes:  Z  X  C  V  B  N  M
        C  D  E  F  G  A  B
```

Key	Note	Key	Note
Z	C	**S**	C#
X	D	**D**	D#
C	E	**G**	F#
V	F	**H**	G#
B	G	**J**	A#
N	A		
M	B		

Press a note key to place that note in the cell under the cursor. The note sounds briefly through the AY so you can hear what you picked.

The note’s **octave** is set by the number keys **1** to **8**. Choose an octave first, then play notes; pressing a number while the cursor sits on an existing note re-tunes that note to the new octave.

Other note-entry keys:

Key	Action
ENTER	Place a **rest** (-=-) – silences the channel from here
SPACE	**Clear** the cell back to empty (---)
I	**Insert** a blank row, pushing later rows down
CAPS SHIFT + 0	**Delete** the row, pulling later rows up
9	**Clear the whole channel** (asks Y/N first)

6.3 Setting volumes, samples and more

The **U** key cycles a small edit mode shown at the top right of the info line, through five settings:

- **Oct** (default) – the number keys set the octave, as above.
- **Vol** – the keys **0–F** set the volume of the cell under the cursor (0 = silent, F = loudest).
- **Smp** – the keys **0–F** set which **sample** (instrument, 00–1F) the cell's note plays with; type two digits for samples above 0F. The chosen number shows in the cell's right-hand **s** column.
- **Orn** – the keys **0–F** set the note's **ornament** (the pitch arpeggio it plays through, 0–F).
- **Noi** – the keys **0–F** set the **master noise pitch** (00–1F) from this row onward. This only takes on a cell that already has a note or rest, because the change rides on an event. The value shows as **Noi:NN**. A *low* number is a bright, hissy noise; a *high* number is a low, coarse rumble. (Each sample can also nudge the pitch per line – see the noise notes under *The instrument editors*.)

Press **U** again to step on; it wraps back round to Oct. (While in any mode other than Oct the letter keys feed the value, so press **U** back round to Oct before using the command keys.)

7. The instrument editors

A *sample* is an instrument: a little per-frame envelope that gives a note its character. An *ornament* is a short pitch arpeggio a note steps through. Press **E** in the pattern view to open the **sample editor**, or **T** for the **ornament editor** (they work the same way).

```
Sample 01  loop 00 len 03
LN b0 b1 b2 b3  Vl Tone TNE Ns
00 FF 8A 00 00  A +    0 TN- 05
01 00 8A 00 00  A +    0 TN- 00
02 00 8A 00 00  A +    0 TN- 00
```

Each sample is a short list of **lines** the AY steps through, one per frame, looping at the **loop** point. Each line is four bytes (**b0–b3**); the editor shows them in hex so every detail of the PT3 format is reachable, with the decoded **Vl** (volume), **Tone** offset, the **TNE** flags and the **Ns** noise pitch alongside:

- **TNE** – whether **t**one, **n**oise and the hardware **e**nvelope are flagged on for that line (a letter when on, **-** when off).
- **Ns** – that line's **noise pitch** (00–1F), or **--** when noise is off on the line. It is added to the song's master noise setting, so a line can ride above or below the master pitch.

7.1 Sample editor keys

- **O / P** – choose which sample (00–1F) or ornament (0–F) to edit.
- **SPACE** – step the field cursor **forward** through the fields.
- **CAPS SHIFT + 5 / 8** – step the field cursor **left / right**.
- **0–F** – set the highlighted byte (two hex digits).
- **T / N** – toggle **tone / noise** on the cursor's line.
- **CAPS SHIFT + 7 / 6** – raise / lower the **noise pitch** (**Ns**) on the cursor's line.
- **len** field – edit it to grow or shrink the instrument (it forks a private copy, so a second instrument won't disturb the first).
- **Q** – return to the pattern view.

The fields the cursor steps through are: **loop**, **length**, then each line's bytes (four per line for a sample, one for an ornament). Setting the **len** field gives the instrument its own private copy, so you can build a second distinct instrument without disturbing the first. Up to 13 lines are shown at once.

For an ornament, each line is a single signed note offset, shown in decimal (e.g. **+0**, **+4**, **+7**).

To use an instrument, set a note's sample (**U-Smp**) or ornament (**U-Orn**) number in the pattern editor, then **A** to hear it.

> **A word on the E (envelope) flag.** **E** shows whether a line is set to use the AY's hardware envelope generator. The flag is shown so that envelopes in songs you *load* survive editing and saving. Authoring a hardware envelope from scratch (choosing its shape and speed) is not in this version – for now, design timbres with the **Vl** column and noise, which covers the vast majority of chip sounds.

8. How instruments make sound

This is the part that takes a moment to click, so it's worth a page.

The AY sound chip is told what to do **50 times a second**. A PT3 song does *not* store those chip settings directly. Instead, every frame the player works them out fresh, per channel, from three things: the **note** you placed, the current line of its **sample**, and the current line of its **ornament**. So you never program the chip directly – you design the sample and ornament tables, and the player "performs" them.

Picture a **sample as a strip of frames** – one line per 1/50th second – that the chip steps through and then loops. Each line says, for that instant: how loud, how far to bend the pitch, whether tone / noise are sounding, and the noise pitch. A short looping strip of those is what gives a note its *character* – a sharp pluck, a steady organ, a noisy hit.

An **ornament** is the same idea for **pitch:** a strip of semitone offsets, one per frame, added to the note. It only moves pitch, not timbre.

8.1 Reading the columns

- **len** – how many lines (frames) the instrument has. **loop** – the line it jumps back to after the last, so it repeats forever from there. (Sustain by looping on a steady line; "play once then go quiet" by looping on a silent line.)
- **LN** – the line (frame) number.
- **Vl** – the **volume** for that frame, **0** (silent) to **F** (loudest). This is the column you'll use most: the shape of **Vl** down the lines *is* the volume envelope.
- **Tone** – a **pitch offset** for that frame (0 = none). Small alternating values make vibrato; a steady ramp makes a pitch bend.
- **TNE** – whether **t**one / **n**oise / hardware **e**nvelope are flagged on this line. Press **T** / **N** to toggle tone / noise on the cursor's line.
- **Ns** – the **noise pitch** for the line (00–1F); **--** when noise is off. Press CAPS+7 / CAPS+6 to tune it.
- **b0 b1 b2 b3** – the raw four bytes, for full control. **Vl** lives in **b1**; **Tone** in **b2/b3**; the tone/noise switches and noise pitch in **b0/b1**. The friendly columns and the T/N/CAPS keys edit these for you, but the hex is always there if you want it.

8.2 Editing a line's volume

The quickest way to shape a sound is its volume over time. The decoded **Vl** column is read from byte **b1**: to set a line's volume, step the field cursor to that line's **b1** and type the volume as the *second* hex digit. For a note that rings cleanly, you want **tone on, noise off** – so first press **T** until the line shows **T**, and **N** until noise shows **-**, then set the volume. (The **T/N** toggles and the volume share the **b1** byte, so set the mixer with **T/N** first, then the level.)

8.3 Making different sounds

Most instruments are just a **volume envelope** (the shape of **Vl** over the lines) plus maybe a little **Tone** movement:

- **Organ / sustained** – several lines all at a steady **Vl** (say **F**), with **loop** on a line that holds. The note rings until the next note.
- **Plucked / decay** – **Vl** starts high and falls, e.g. **F C 9 6 3 0**; loop the last (silent) line so it stays quiet. Short + sharp = a blip; longer = a mellow pluck.
- **Bass** – a low note with a steady mid **Vl**, kept short and looping.
- **Vibrato** – hold **Vl** steady and put small alternating **Tone** values on successive lines (e.g. **+8, 0, -8, 0 …**).
- **Chords / arpeggios** – this is an **ornament**, not a sample. An ornament of **0, 4, 7** looping every three frames turns one held note into a major

chord shimmer (the classic chiptune sound); **0, 3, 7** = minor, **0, 12** = octave. Assign it to a note with U-Orn.
- **Percussion / snare / hi-hat** – these need **noise** instead of (or as well as) tone, plus a fast Vl decay. Press N to turn noise on (the TNE column shows N), set Vl to fall quickly (e.g. **F 9 4 0**), and you have a hit. Press T to drop the tone for pure noise. **Tune the noise** two ways: per line with CAPS+7 / CAPS+6 in the sample editor (the **Ns** column) so a hit can sweep bright-to-dark, and per song with U-Noi in the pattern editor to set the master pitch from a row onward. A bright **Ns** (low number) gives a hi-hat or snare crack; a dark **Ns** (high number) gives a tom or rumble.

8.4 Where it goes

Each frame the player turns this data into the AY's registers: note + ornament set the channel's **pitch**; the sample's **Tone** is added to that pitch; Vl becomes the channel **volume**; the tone/noise switches set the chip's **mixer**; and the noise pitch sets the shared noise generator. All of it is stored inside the song file (the sample and ornament tables, and the noise settings), so it travels with the tune when you save.

9. Hearing your work

You don't need to leave the editor to listen:

Key	Action
A	**Play the whole song** from the current pattern
L	**Loop the current pattern** over and over

Either way, press any key to stop and return to editing. TS Tracker rebuilds the tune from your edits before it plays, so you always hear the latest version – including the tempo you set on the Song info screen.

10. Saving to tape

Press W to save. TS Tracker first rebuilds your song into a proper PT3 file, then asks for a name:

```
Filename (max 8 chars).
Letters / digits / space.
    MY TUNE  01
Bytes:   234
```

Type up to eight characters. TS Tracker adds a two-digit version number automatically and bumps it on every save, so MY TUNE 01, MY TUNE 02, and so on never overwrite each other on the tape.

Key	Action
DEL (CAPS SHIFT + 0)	Erase the last character
ENTER	Start recording - have the tape ready first!
Q	Cancel

Start your recorder, press ENTER, and the song is written as a standard CODE block. Load it back with the TS Tracker Player, or scan it straight back into the editor.

> If the song has grown too large to fit the cassette song area, TS Tracker refuses to save and tells you so. Remove some notes or patterns and try again.

11. Walkthrough: your first tune

Theory is fine, but the keys live in your fingers only after you use them. Here is a complete pass - a short loop with a melody, a bass and a drum - from a blank machine to a saved tape. Follow along.

1. Start a new song. From the first screen press N. You land on the Song info screen; press V to open the pattern editor on an empty pattern 0. The cursor sits on row 00, channel A.

2. Lay a melody on channel A. Octave 4 is fine (the top line shows Oct:4). Place a note every four rows so they fall on the cyan beat lines:

- Row 00: tap Z (C).
- Press CAPS+6 to row 04: tap B (G).
- Down to row 08: tap N (A).
- Down to row 0C (the next bar line): tap M (B).

You have placed four notes, one per beat. Press A to hear the loop; press any key to stop.

3. Add a bass on channel B. Press CAPS+8 to move to channel B, then R to jump back to row 00. Press 2 to drop to octave 2, then:

- Row 00: Z (a low C).
- Down to row 08: V (a low F).

Press A again - melody and bass together.

4. Add a drum on channel C. CAPS+8 to channel C, R to row 00. Place a hit on the off-beats:

- Row 04: tap any note (say Z) - we will give it a noise instrument.

- Down to row 0C: Z again.

5. Build a snare instrument. Press E for the sample editor. Press P until you are on a free sample, say 02. Step to the **len** field (SPACE twice, or CAPS+5/8) and type 4 for four lines. Now make a noisy decay:

- On each line, press N until the TNE column shows N, and T until tone shows - (pure noise).
- Set the volumes to fall fast: step to line 00's **b1** and type the level in the second digit so Vl reads F; line 01 9; line 02 4; line 03 0. The Vl column now reads F 9 4 0.
- Tune the crack: with the cursor on a line, tap CAPS+7 / CAPS+6 to set the Ns noise pitch to taste (a low number for a sharp snare).
- Set **loop** to 03 (the silent last line) so the hit fires once and stops.

Press Q to return to the pattern editor.

6. Point the drum notes at the snare. You are on channel C. Press U until the mode reads **Smp:**, then type 02 to set this cell's sample to your snare. Move to the other drum row (CAPS+6 to row 0C) and do the same. Press U back round to **Oct:** so the letter keys are commands again.

7. Set the tempo. Press Q to the Song info screen. Press CAPS+7 / CAPS+6 to set **Speed** - try 5. Press V to return to the pattern.

8. Play the whole thing. Press A. Press any key to stop.

9. Save it. Press W, type a name like **FIRST**, make sure your recorder is in record, and press ENTER. TS Tracker writes **FIRST 01** to the tape.

That's a finished, saved chip-tune - three voices, a custom drum, and a tempo of your own. Everything else in this manual is variations on those steps.

12. The help screen

Press K at any time in the pattern editor for a full one-screen key reference. Press any key to return exactly where you left off.

13. Quick key reference

First screen: S scan, N new song, Q quit.

Directory: 1-9 choose, R re-scan, Q quit.

Song info: V edit patterns, CAPS+7/6 Speed, Q back.

Pattern editor:

Key	Action	Key	Action
CAPS+5/6/7/8	Move cursor	Z..M	Play notes
O / P	Prev / next pattern	1..8	Octave
F	Jump to pattern	ENTER	Rest
R	Jump to row 0	SPACE	Clear cell
A	Play song	I	Insert row
L	Loop pattern	CAPS+0	Delete row
U	Oct/Vol/Smp/Orn/Noi	9	Clear channel
E	Sample editor	T	Ornament editor
W	Save to tape	K	Help
Q	Back to song info		

Sample / ornament editor:

Key	Action
O / P	Choose instrument
SPACE, CAPS+5/8	Move between fields
0-F	Set the byte (hex)
T / N	Toggle tone / noise (samples)
CAPS+7/6	Noise pitch up / down (samples)
Q	Back to pattern editor

14. Limits and notes

- A song may use up to **14 patterns**.
- When you save, the whole tune must fit the cassette song area (about **7 K**, instruments + patterns combined). The **Free** counter warns you as you approach the limit.
- TS Tracker edits **PT3** songs. **PT2** songs play but cannot be edited.
- The master noise pitch (**U-Noi**) and the per-line noise pitch (**Ns**) attach to a row that carries a note or rest - they ride on an event, the way the PT3 format expects.
- An empty first row on a sounding channel is saved as a rest - silence at the start of the pattern - because the PT3 format always reads row zero.
- The sample and ornament editors show up to 13 lines of an instrument at once.
- Authoring a hardware envelope (its shape and speed) is not in this version; the **E** flag is shown so loaded songs keep theirs.

15. Credits

TS Tracker is a 64K Software production for the Timex Sinclair 2068. It uses the PTxPlay (Bulba) AY replay driver, shared with the TS Tracker Player. The display style follows the period TS2068 convention of the TIMEX banner, status bar and inverse-video hot-keys.

Happy composing.

TS Tracker - a 64K Software production - End of manual